PROJECT REQUIREMENTS DOCUMENT

WEEK 3 — TASK | DATA ANALYSIS

GOOGLE MAPS BUSINESS ANALYSIS

Find. Analyze. Recommend.

A structured plan to identify local businesses without websites, analyze their digital presence, and recommend practical digital-growth opportunities.

Document purpose

This document converts the supplied task poster into an actionable project specification. It defines the objective, scope, workflow, data fields, analysis, tools, deliverables, and submission checks for the project.

Project outcome

A verified dataset and a clear, evidence-based shortlist of local businesses that may benefit from establishing or improving a website and their broader online presence.

Prepared from the project brief shown in the supplied image. Project owner, institution, location, and deadline were not specified, so these are intentionally left open.

1. Project Overview

Project title: Google Maps Business Analysis

Task label: Week 3 — Data Analysis

Core idea: Analyze local businesses listed on Google Maps, identify businesses that do not appear to have a website, compare their digital presence, and recommend realistic opportunities for digital growth.

1.1 Problem statement

Many local businesses may depend on map listings, phone calls, social pages, or walk-in customers without having a dedicated website. This can make it harder for potential customers to find complete information, explore services, build trust, or contact the business. The project will use structured research and analysis to find such businesses and prioritize useful recommendations.

1.2 Objectives

- Collect a structured list of local businesses from Google Maps and record their available public listing information.
- Identify whether each business has a discoverable official website, using a consistent verification process.
- Compare businesses by category, rating, review count, location, and website availability.
- Visualize the distribution of businesses with and without websites.
- Identify high-potential businesses for digital improvement using transparent criteria.
- Prepare a ranked top-10 business list and practical recommendations supported by the collected evidence.

1.3 Expected users and use

The primary audience is the project evaluator or instructor. The analysis may also be useful to a local-business support team, digital marketing student, or small-business consultant who wants to understand gaps in online presence.

2. Scope and Boundaries

Included in scope

- A defined local area or set of nearby areas, selected before data collection.
- Public business listing information visible on Google Maps or other permitted public sources.
- Website presence checks and a documented method for marking website status.
- Data cleaning, category-wise and location-wise comparisons, charts, rankings, and recommendations.
- Deliverables in Excel/CSV, charts or dashboard, presentation, top-10 list, and final recommendations.

Not included unless the instructor separately approves

- Building or deploying websites for the businesses.
- Contacting businesses, running paid campaigns, or promising business results.
- Collecting private, login-protected, or sensitive personal data.
- Bypassing website restrictions or scraping in ways that violate platform terms or applicable law.
- Treating an absent website link as absolute proof that a business has no website without verification.

Important definition

“No website” should mean “no official website found after the defined checks.” A missing website field on a map listing is a lead for verification, not conclusive proof. Record the verification method and status so that the analysis remains fair and reproducible.

3. Functional Requirements

ID	Requirement	Description
FR-01	Define study area	Record the city, locality, radius, or list of neighborhoods covered by the study.
FR-02	Collect businesses	Build a list of relevant businesses from public Google Maps listings.
FR-03	Capture listing details	Record name, category, address/locality, rating, review count, phone if publicly listed, and website link/status when available.
FR-04	Verify website status	Check the listing website field and perform a reasonable official-site search; record Found / Not Found / Unclear.
FR-05	Clean data	Remove duplicates, standardize categories and text, handle missing values, and validate numeric fields.
FR-06	Analyze dataset	Compare website status by category and area; summarize ratings, review counts, and opportunity indicators.
FR-07	Visualize findings	Create charts and/or a dashboard that clearly communicate the main patterns.
FR-08	Rank opportunities	Create a transparent shortlist and top-10 list with a short reason for each selection.
FR-09	Recommend actions	Suggest relevant digital improvements, such as a basic website, contact page, service pages, local SEO, or clearer business information.
FR-10	Export deliverables	Save a clean Excel/CSV dataset, charts/dashboard, presentation, and recommendation list.

3.1 Non-functional requirements

- **Accuracy:** Every record should be traceable to a public source and checked for obvious errors.
- **Consistency:** Use the same definitions and verification method for every business.
- **Readability:** Charts, tables, and slides should have clear titles, labels, units, and sources.
- **Reproducibility:** Document the study area, date of collection, search terms, and decision rules.
- **Responsible data handling:** Use only necessary public business information and respect platform terms and applicable law.

4. Data Collection and Dataset Design

4.1 Suggested data fields

Field	Definition / rule
Business ID	Unique identifier created for each row, e.g., B001.
Business Name	Name displayed on the public listing.
Primary Category	Business type, such as café, salon, clinic, repair shop, or retailer.
Address / Locality	Public business address or neighborhood; standardize spelling.
Google Maps Listing URL	Link to the listing, where available and permitted.
Rating	Public average rating; leave blank if unavailable.
Review Count	Public number of reviews; store as a number.

Field	Definition / rule
Public Phone	Public business contact number, if needed for the project.
Website URL	Official website link if found; do not label social media as a website.
Website Status	Found / Not Found / Unclear.
Verification Notes	Brief note about how the website status was checked.
Collection Date	Date the information was observed.
Opportunity Flag	Yes / No / Review, based on the documented scoring rules.
Recommendation	Short action relevant to the business and evidence.

4.2 Data collection procedure

1. Choose and write down the study area and business categories before collecting records.
2. Search Google Maps using consistent category and locality search terms.
3. Record the public listing details needed for the analysis and retain the listing reference where possible.
4. Check the website field on the listing. If it is missing, search for an official website using the business name and locality.
5. Mark status as Found, Not Found, or Unclear. Add a short verification note, especially for ambiguous names.
6. Record the date of observation because ratings, reviews, hours, and links can change.
7. Review a sample of records a second time to catch inconsistent classifications.

4.3 Data quality checks

- Business name and locality are present for each included record.
- No accidental duplicate rows for the same business and location.
- Ratings fall within the displayed rating scale; review counts are non-negative whole numbers.
- Website links are validly formatted and appear to belong to the business.
- Missing or ambiguous values are labelled rather than guessed.
- Website status and opportunity labels follow one consistent rule.

5. Analysis and Opportunity Identification

5.1 Required analysis

- Total number of businesses researched.
- Number and percentage with a website versus no website found.
- Category-wise comparison of website presence.
- Location-wise or locality-wise comparison of website presence.
- Rating and review-count summaries for businesses with and without websites, where data is sufficient.
- A shortlist of businesses that may benefit from improved digital presence.

5.2 Suggested charts

Question	Visual	Purpose
Website status	Doughnut or pie chart	Share of businesses with a website, without a website found, or unclear.
Website status by category	Stacked bar chart	Compare digital presence across business categories.
Businesses by locality	Bar chart or map	Show where the researched businesses are concentrated.
Rating vs. review count	Scatter plot	Explore listing engagement patterns; avoid implying causation.
Top opportunities	Ranked bar/table	Show selected businesses and the evidence behind their ranking.

5.3 Transparent opportunity scoring

Use a simple rubric to prioritize follow-up. The score is a project screening tool, not proof that a business will gain customers or revenue.

Factor	Points	Scoring guidance
No official website found after checks	0–3	3 = not found; 1 = unclear; 0 = official website found.
Public rating	0–2	Use a consistent rating band; explain the selected cutoffs.
Review count	0–2	Higher review activity may indicate an established customer base; set bands before scoring.
Business information completeness	0–1	Award point when key listing details are present and usable.
Clear improvement opportunity	0–2	Evidence-based gap such as missing service details, booking information, menu, or contact page.

Maximum suggested score: 10. Define the rating and review-count bands before scoring, apply them consistently, and include the score rationale beside each shortlisted business. Do not infer that a business is successful or unsuccessful from rating alone.

6. Recommended Technology Stack

Layer	Tools / use
Data collection	Google Maps public listings and manual research; use only permitted methods.
Data handling	Microsoft Excel or Google Sheets for tabular collection, cleaning, filters, and validation.
Analysis	Excel formulas and pivot tables; optional Python with Pandas for larger datasets.
Visualization	Excel charts or Power BI / Tableau; optional Python Matplotlib or Seaborn if permitted by course requirements.
Programming (optional)	Python with Pandas and Matplotlib for repeatable cleaning and analysis.
Presentation	PowerPoint summarizing the objective, method, key findings, charts, top-10 list, and recommendations.

The image lists Excel / Google Sheets for data handling, Power BI / Tableau / Excel charts for analysis and visualization, and optional Python tools including Pandas, Matplotlib, and Seaborn for advanced analysis.

7. Deliverables

Deliverable	Acceptance criteria
1. Dataset (Excel/CSV)	Clean table of researched businesses with website status, source references, collection date, and verification notes.
2. Dashboard / Charts	Clear visual summary of website presence and category/location patterns.
3. PPT (Insights & Opportunities)	Presentation explaining objective, data method, key findings, visuals, limitations, and recommendations.
4. Top 10 Business List	Ranked list of ten priority businesses, where the dataset contains enough valid candidates, with rationale for each.
5. Final Recommendations	Practical digital-growth actions linked to the observed gaps and evidence.

If fewer than ten valid candidates are found, report the actual number and explain why rather than filling the list with unsupported entries.

8. Suggested Project Workflow

1. Plan	Choose study area, categories, collection date, and definitions.
2. Collect	Build the initial business dataset from public listings.
3. Verify	Check website status and add verification notes.
4. Clean	Remove duplicates, standardize fields, and validate values.
5. Analyze	Calculate counts, percentages, category/location comparisons, and opportunity scores.
6. Visualize	Create charts and a dashboard with titles and source notes.
7. Recommend	Rank opportunities and write specific, realistic actions.
8. Submit	Check the dataset, charts, PPT, top-10 list, and final recommendations.

9. Submission Guidelines and Evaluation Readiness

The task poster specifies the following submission sequence:

1. Collect and analyze data.
2. Prepare a dashboard and insights.
3. Add top opportunities.
4. Submit the Excel file and PPT.
5. Marks will be out of 10.

Suggested self-check before submission

Check	Pass condition
Dataset	File opens correctly; columns are named; duplicate and missing values are handled.
Evidence	Website-status decisions are documented; listing links or source notes are included.
Analysis	Counts and percentages match the dataset; charts use correct labels and totals.
Recommendations	Each recommendation explains the observed gap and a plausible next step.

Check	Pass condition
Presentation	PPT is readable, concise, and includes method, findings, limitations, and conclusion.
Ethics	Only appropriate public information is used; platform rules and applicable laws are respected.

The exact grading rubric, required number of businesses, target locality, and submission deadline were not visible in the supplied poster. Confirm these with the instructor and update this document if necessary.

10. Risks, Assumptions, and Limitations

Assumptions

- The project is an academic data-analysis task and not a commercial lead-generation campaign.
- The study area and categories will be selected by the student or instructor.
- Excel/CSV and a PowerPoint presentation are expected output formats, as indicated in the task poster.
- The project can use manual research and spreadsheet analysis; Python is optional.

Risks and mitigations

Risk	Mitigation
Incorrect website classification	Use Found / Not Found / Unclear, verify missing links, and retain notes.
Outdated or changing listings	Record the collection date and avoid presenting data as permanently current.
Duplicate or inconsistent data	Use unique IDs, standardized categories, and duplicate checks.
Biased opportunity ranking	Publish scoring rules and apply them consistently.
Platform restrictions	Avoid prohibited scraping or automated access; follow Google Maps terms and applicable law.
Overclaiming results	Describe opportunities as hypotheses for improvement, not guaranteed outcomes.

Limitations

A missing website link does not prove that a business lacks a website. Ratings and review counts do not measure every aspect of business quality. Public listing data may be incomplete or change after collection. The results represent the selected study area and sample, not all local businesses.

11. Definition of Done

The project is ready to submit when all of the following are true:

- The study area, sample, and collection method are documented.
- The dataset is cleaned and contains source/verification notes.
- Website status is classified consistently and uncertain cases are clearly labelled.
- Required comparisons and charts are complete and agree with the dataset.
- A top-10 list or smaller evidence-based shortlist is included.
- Recommendations are specific, practical, and tied to findings.
- Excel/CSV and PPT files open correctly and are ready to submit.

Project summary

Collect public local-business data → verify website presence → clean and analyze the dataset → visualize patterns → prioritize businesses → recommend digital-growth actions → submit the dataset and presentation.

End of document